GHANGE NOTICES ARE NOT CUMULATIVE AND SHALL BE RETAINED UNTIL SUCH TIME AS THE ENTIRE STANDARD IS REVISED.

Fed. Std. No. 162a
May 9, 1967
Change Notice 1
January 4, 1968

FEDERAL STANDARD

BOSE, RUBBER, VISUAL INSPECTION GUIDE FOR

The following changes in Fed. Std. No. 162a, dated May 9, 1967, have been approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. Corrected table and figures. The following corrections are a part of this standard:

CONTENTS

Number	Title	Pages
Table I	Clarification of defects	9, 10
Figure:		
30	Depression, small and light: class A—major; classes B and C—minor	51
31	Depression, small and light: class A—major; classes B and C—minor	52
32	Depression, small and light: class A—major; classes B and C—minor	53
33	Depression, small and light—minor	54
42	Lap : foldover, opening—major	63
43	Lap: foldover, opening—major	64
48	Charge mark: I—major. II—minor. III—Not a defect	69
49	Charge mark: class A—minor; classes B and C—not a defect	70
62	Light cover (roughness), thickness not below specified or not specified—minor	83
63	Light cover (roughness), thickness not below specified or not specified—minor	84
72	Patched cover, open edge—major	93
73	Patched cover: I—Incorrect thickness, high or low—major. II—Correct thickness and securely cured—not a defect	94
76	Porosity, small localized area, reinforcement not exposed minor	97
77	Pricker marks through to reinforcement—major (see note in table I for other classification)	98
90	Wrinkles, wrapper, medium—major	111
91	Wrinkles, wrapper, small—minor	112
110	Gasket, tail or binder washer for expansion ring coupling: I—Not properly placed—major. II—Correct placement	131
111	Insertion distance wrong—major	132
112	Ring, expansion, improperly placed: I—Correct. II—Ring in waterway—major. III—Too much gap—major	133
113	Cement in tube—minor	134
116	Clamp mark—major	137

RETAIN THIS COVER PAGE AND INSERT BEFORE THE TABLE OF CONTENTS OF THIS STANDARD

FSC 4720

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

Number	Title	Pages
Figure:		
117	Clamp mark—major	138
130	Ball—up: I—Yarm ball up—major. II—Nomal hose	151
131	Ball-up—major	152
138	Friction between braids, where specified, missing: I—Friction missing—major. II—Normal hose	159
139	Knot in yarn: I—Knot in yarn—major. II—Normal hose	160
152	Wire through cover—major	173
153	Wire through tube—major	174
166	Wire, static, bulge: with loose cover—major; dynamic use (Air Force procurement only)—major; with adhered cover—not a defect	187
167	Yarn through cover—major	188
174	Hole—major	195
175	Hole—major	196
186	Tube missing: I—Normal hose. II—Major	207
187	Off—center tube: I—Thickness meets tolerance—not a defect. II—Thickness below tolerance—major	203
192	Porosity, more than 15 percent of tube thickness—major	213
193	Ring: class A—major; classes B and C—not a defect	214
200	Surface irregularity: when assembled with swaged fittings—major; all others—minor	221
201	Surface irregularity: when assembled with swaged fittings—major; all others—minor	222
202	Surface irregularity: when assembled with swaged fittings—major; all others—minor	223
203	Wire through tube—major	224

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

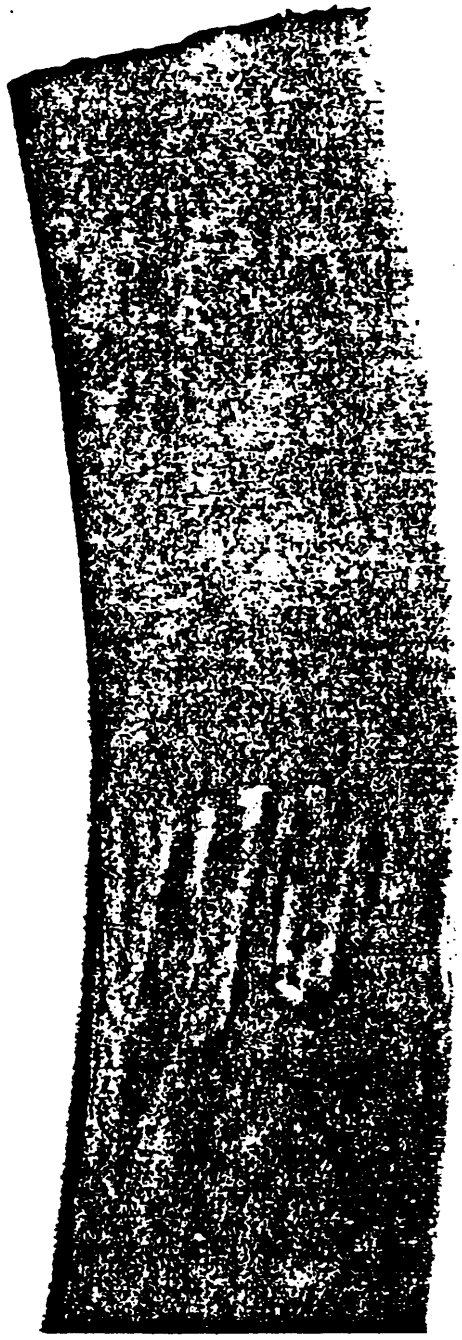


FIGURE 30. Depression, small ad light: class A—major; classes B and C—minor. (Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

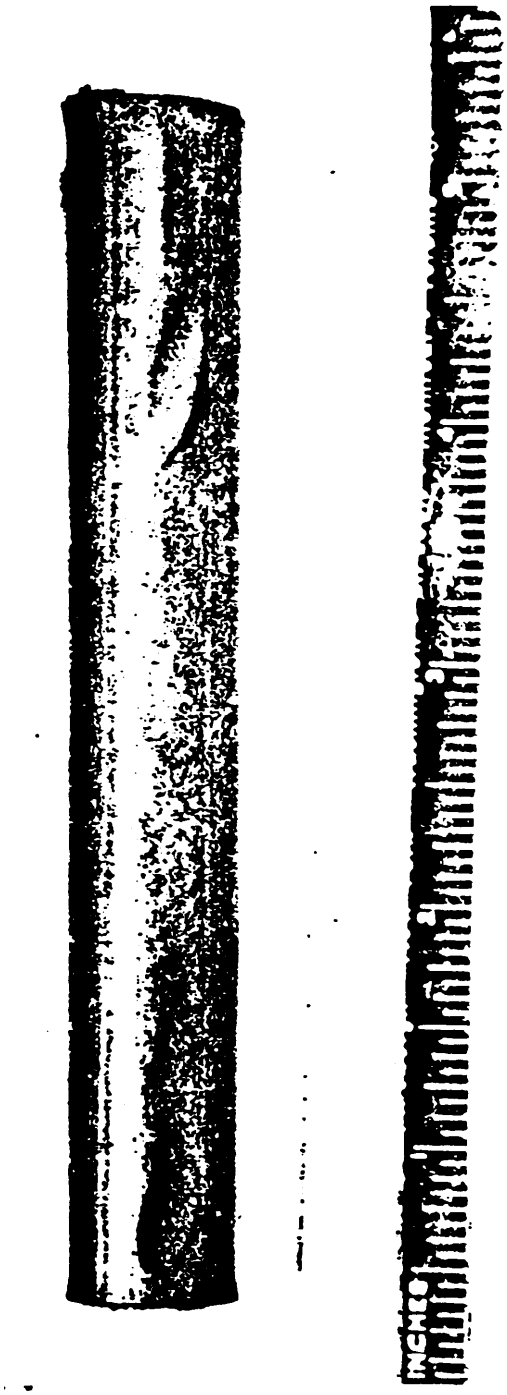


FIGURE 31. Depression, small and light: class A—major: classes B and C—minor. (Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

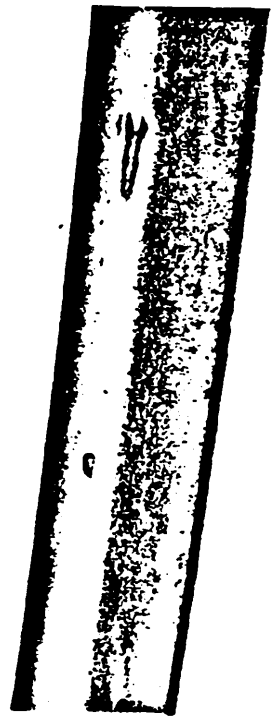


FIGURE 32. Depression. small and light: class A—major: classes B and C—minor.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

DELETED

FIGURE 33. Depression, small and light—minor
(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

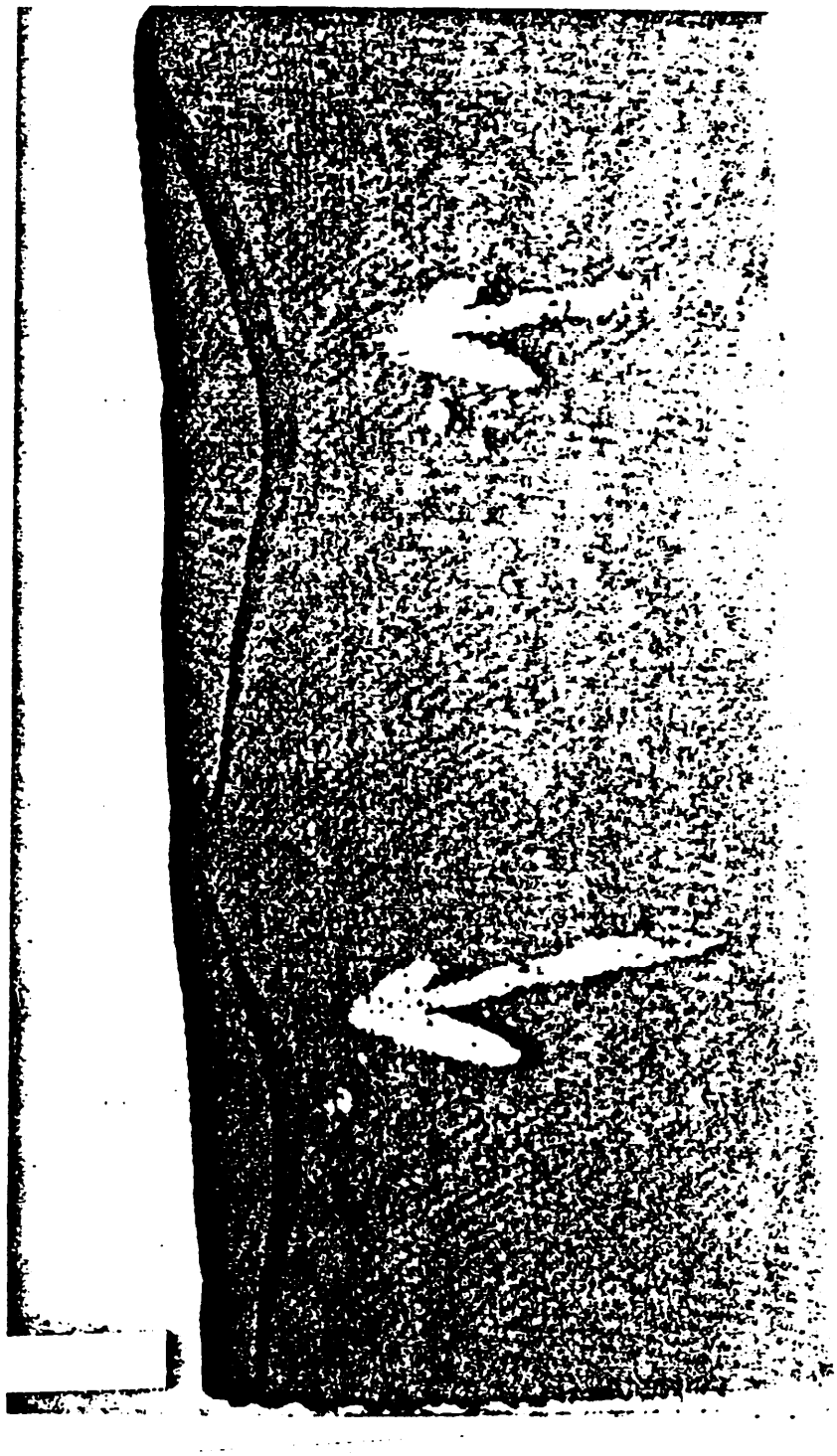


FIGURE 42. Lap: foldover, opening—major,
(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

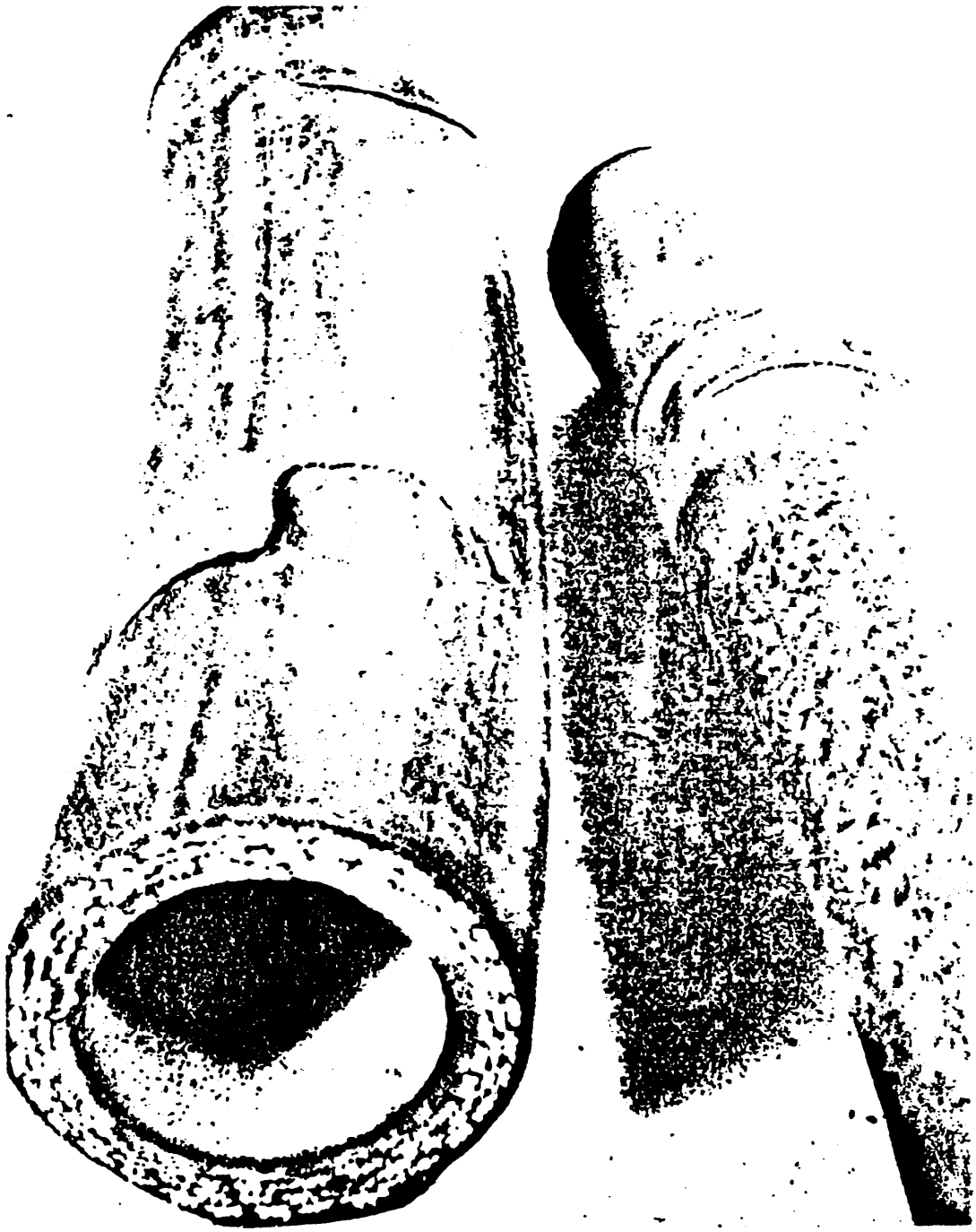


FIGURE 43. Lap: folkover, opening—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.



FIGURE 48. Charge mark:

I—major.
II—minor.
III—Not a defect.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

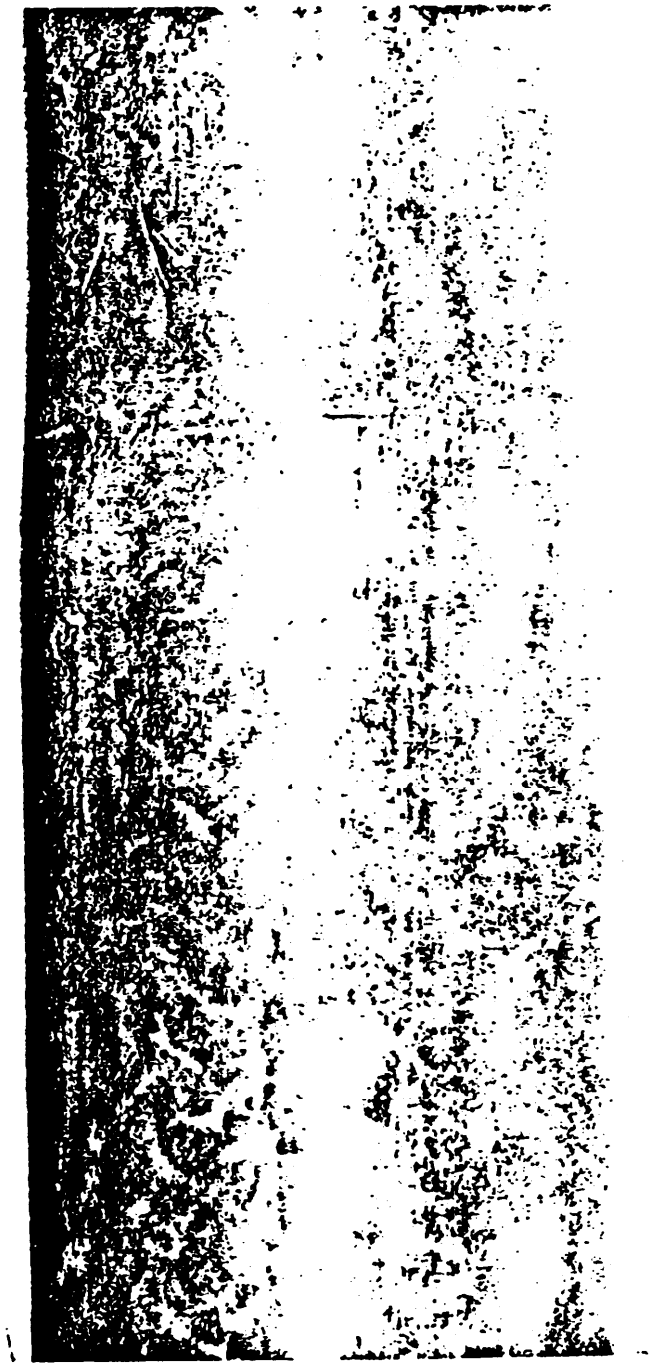


FIGURE 49. Charge mark: class A—minor; classes B and C—not a defect. (Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.



FIGURE 62. Light cover (roughness), thickness not below specified or not specified —minor.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

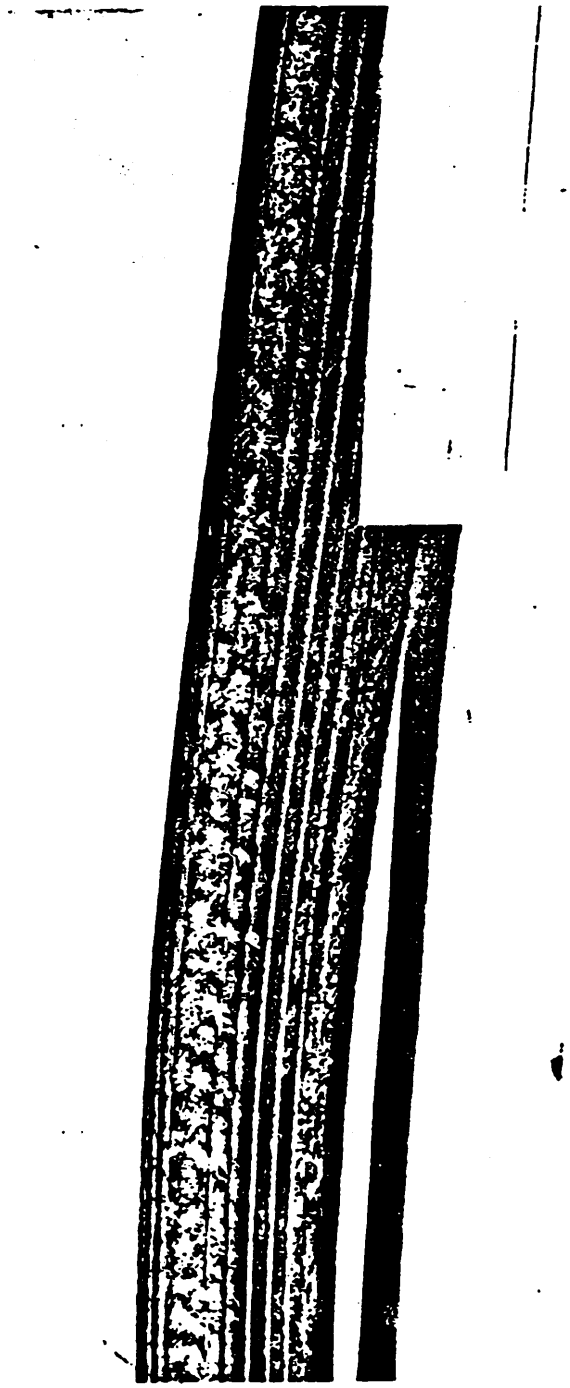


FIGURE 63. Light cover (roughness), thickness not below specified or not specified —minor.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

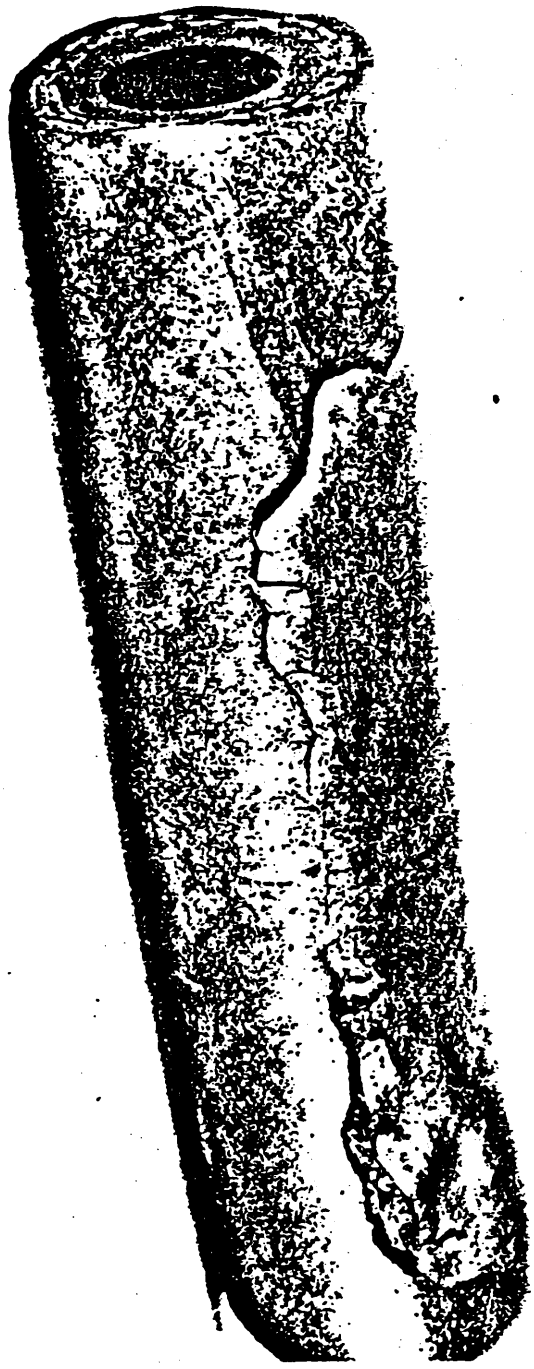


FIGURE 72. Patched cover, open edge—major.
(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

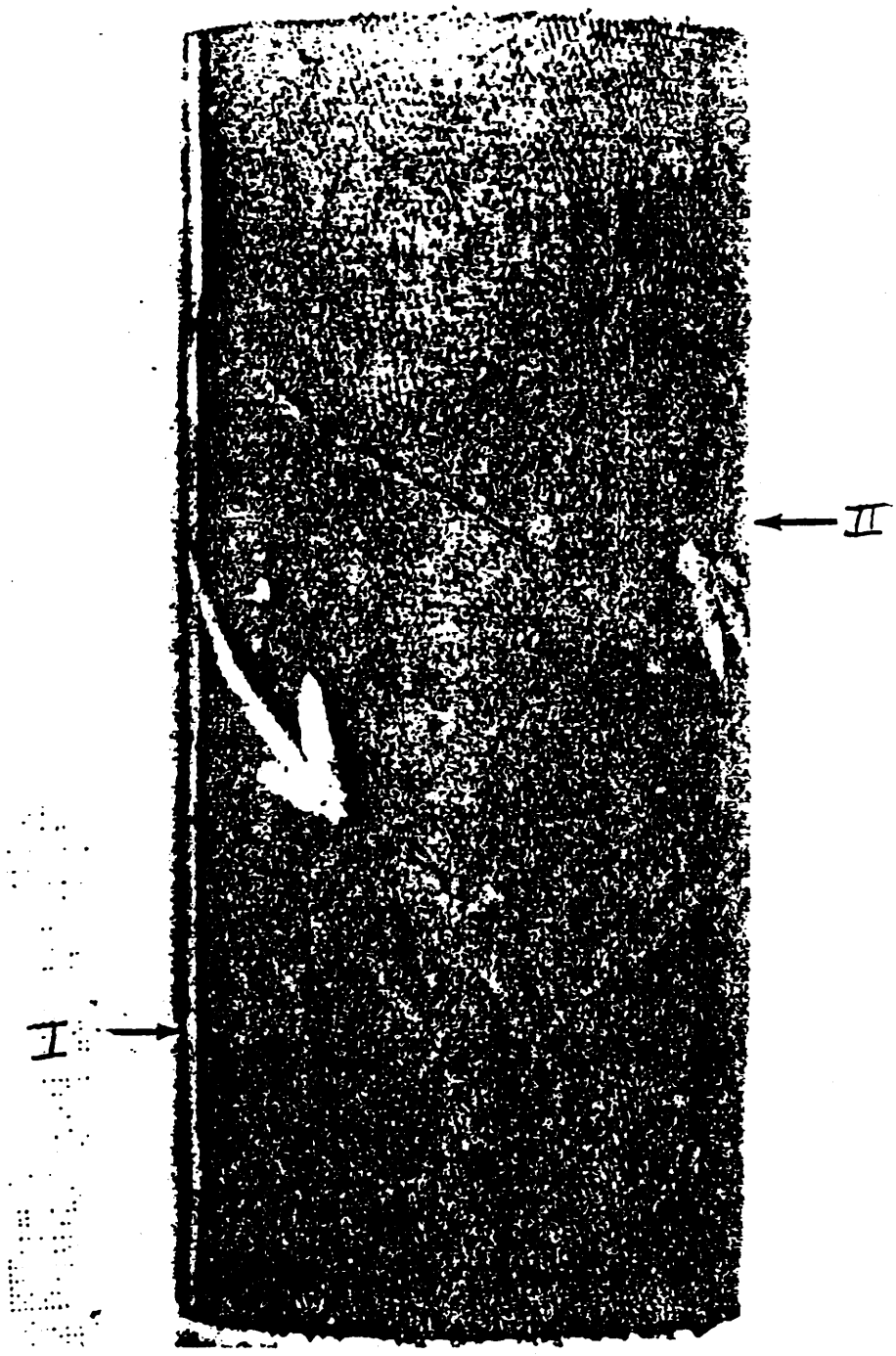


FIGURE 73. Patched cover:

I—Incorrect thickness, high or low—major.
II—Correct thickness and securely cured—not a defect.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

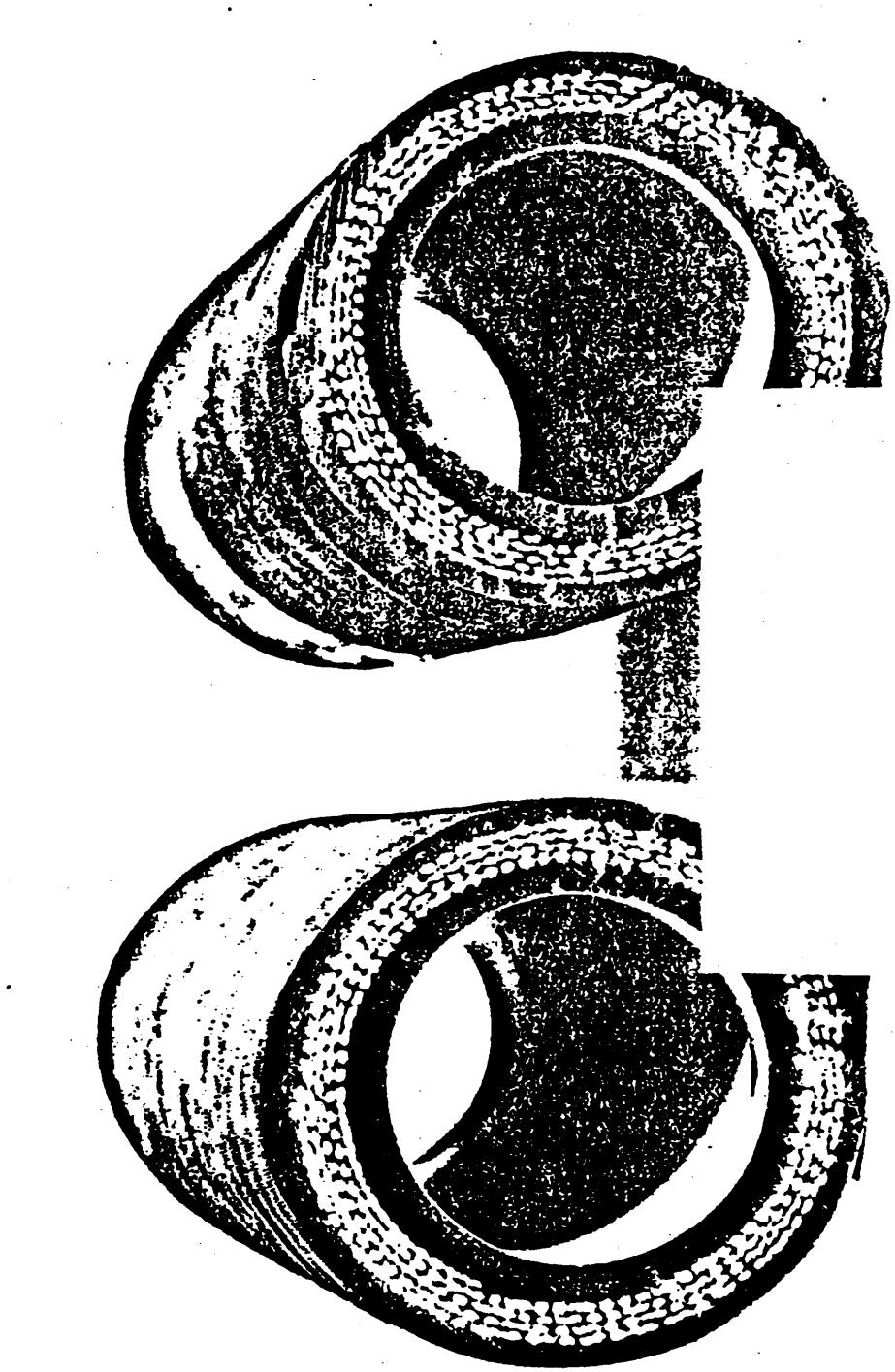


FIGURE 76. Porosity, small localized area, reinforcement not exposed—minor.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

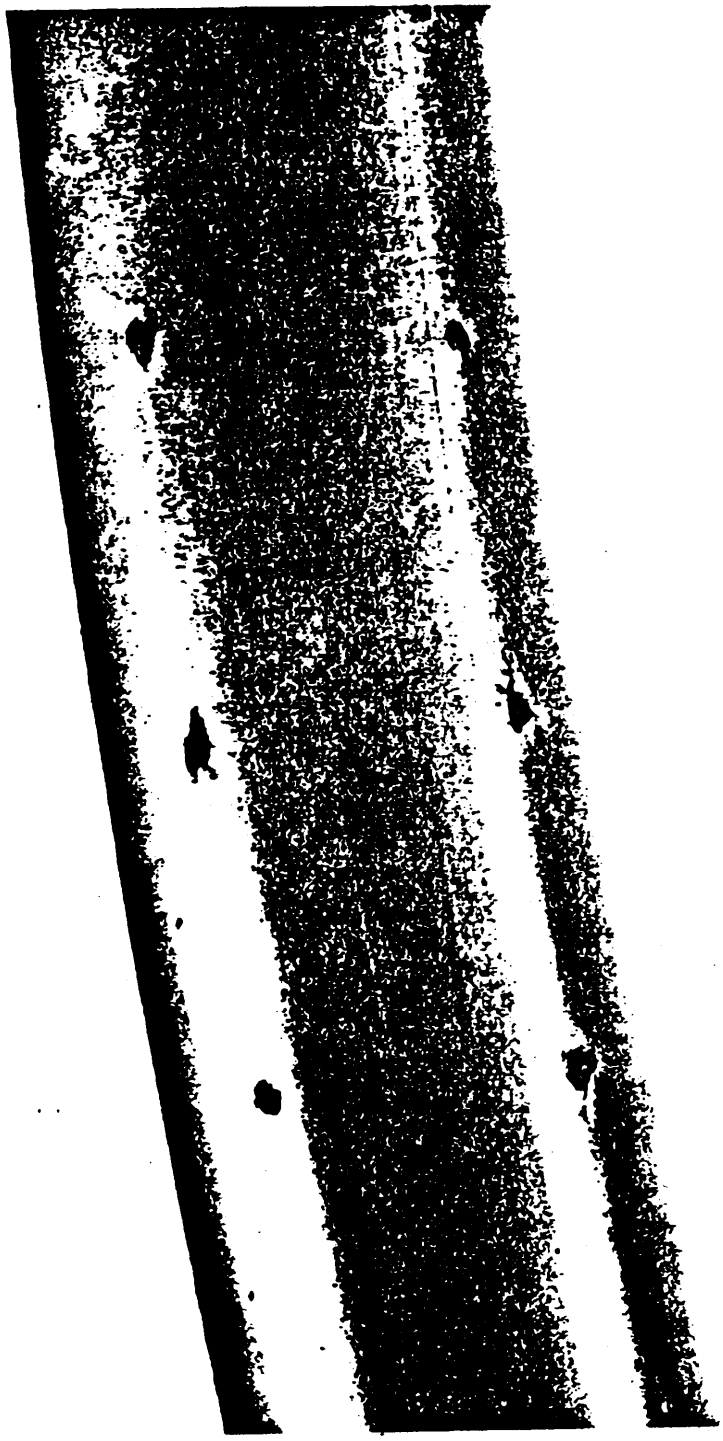


FIGURE 77. Pricker marks through to reinforcement—major (see note in table I for other classification).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

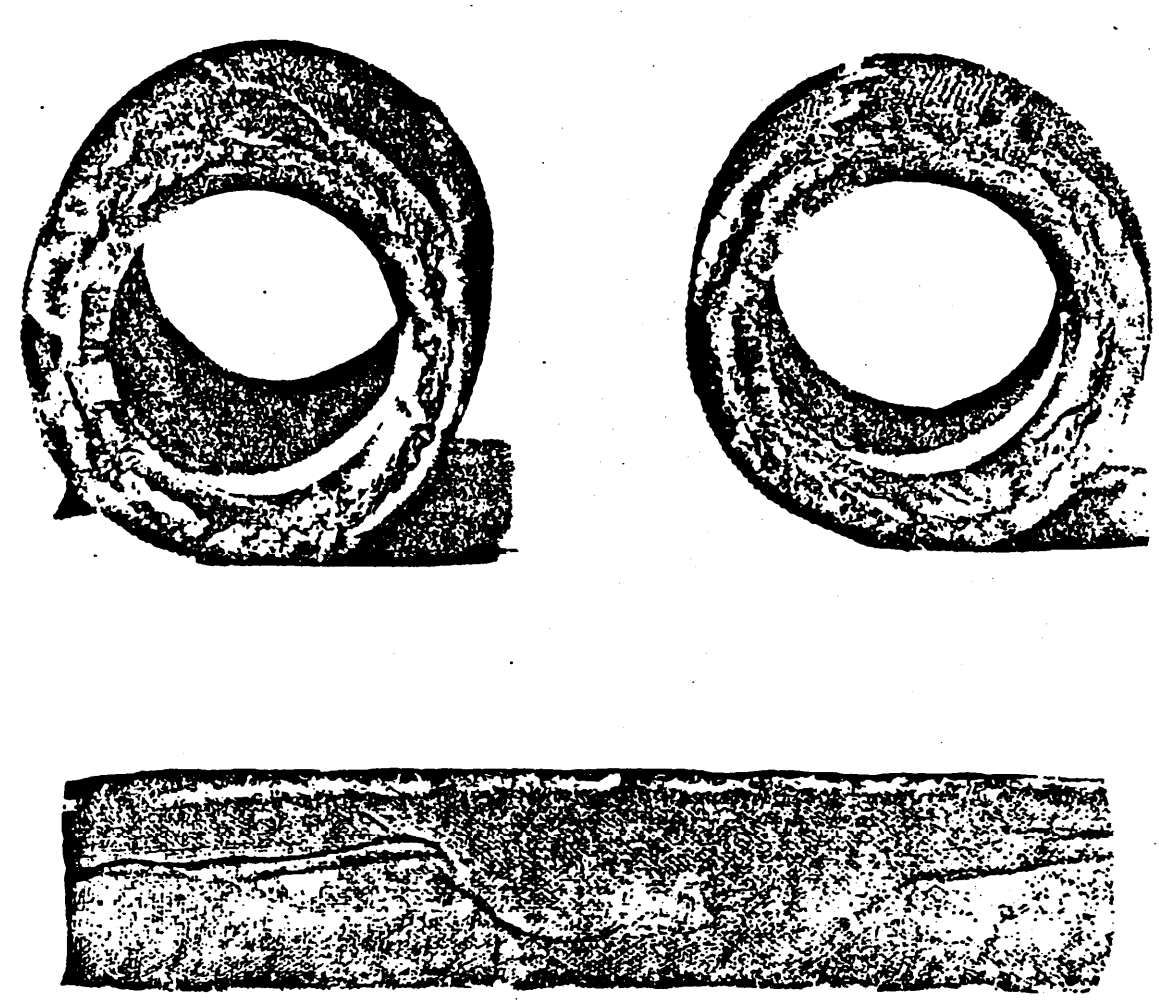


FIGURE 90. Wrinkles, wrapper, medium—major.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

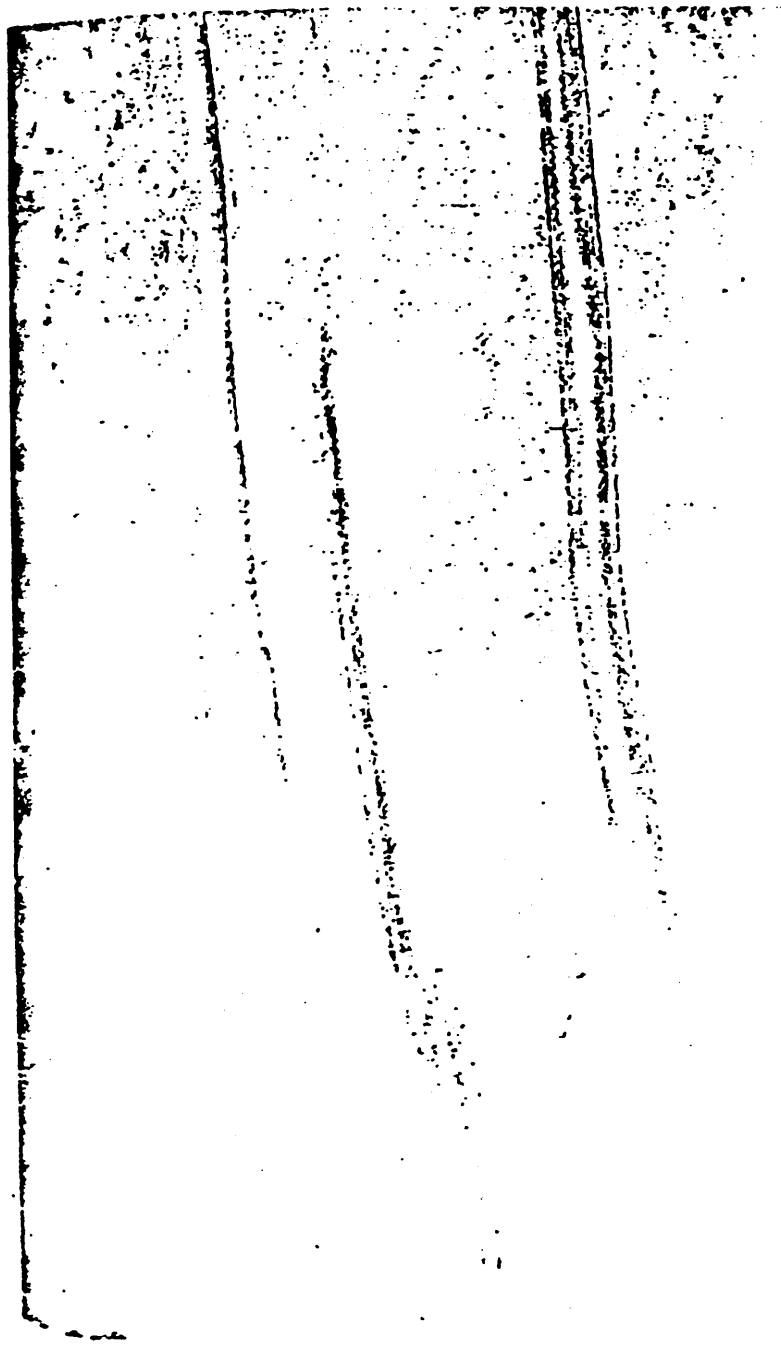


FIGURE 91. Wrinkles, wrapper, small—minor.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

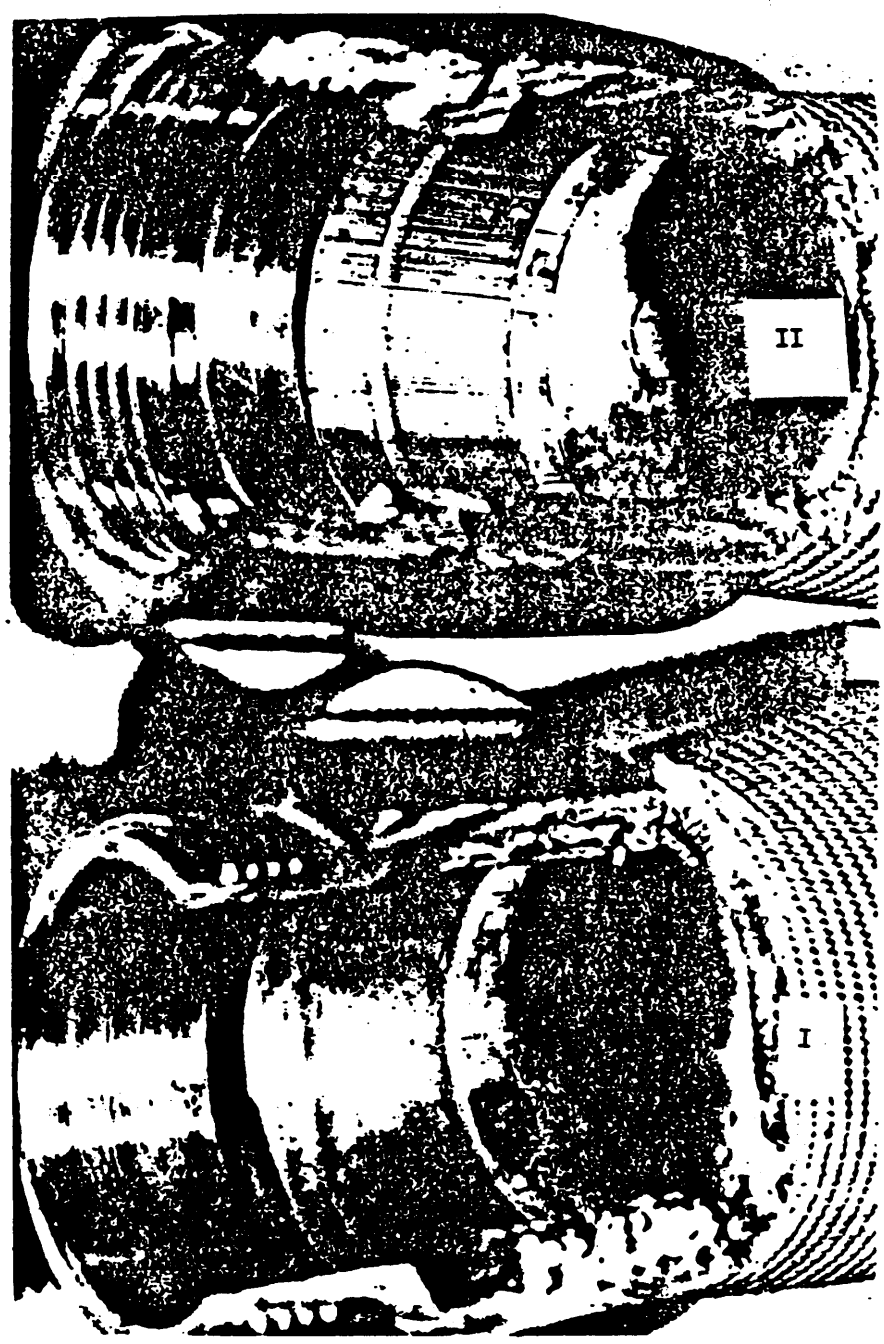


FIGURE 110. Gasket, tail or binder washer for expansion ring coupling:
I—Not properly placed—major.
II—Correct placement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

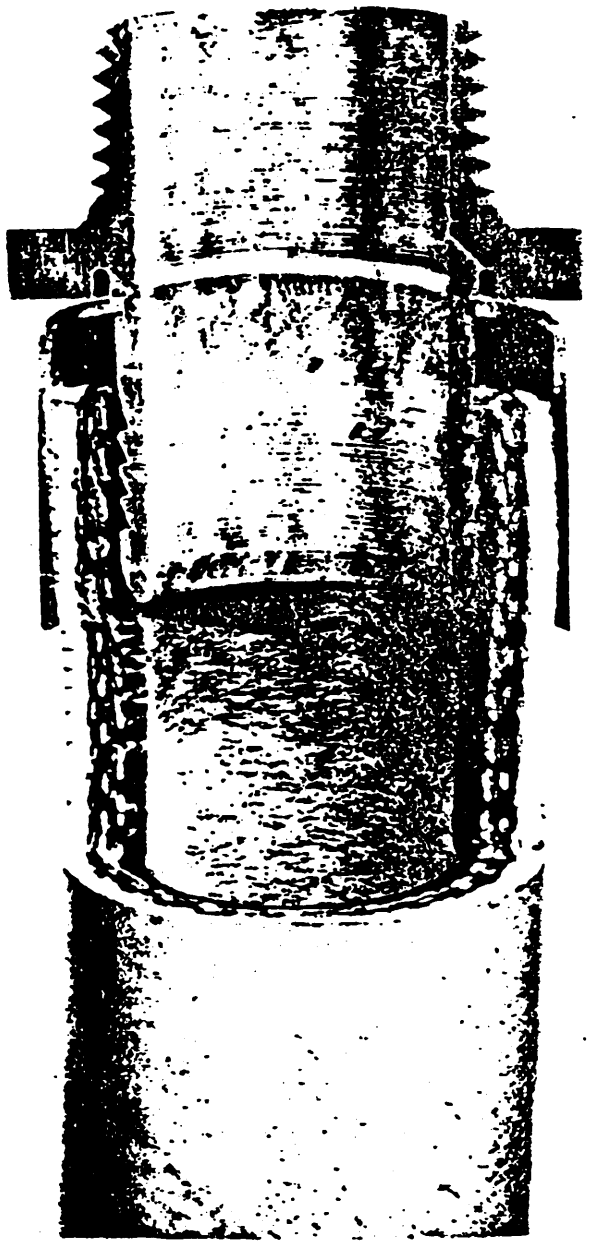


FIGURE 111. Insertion distance wrong—major.
(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

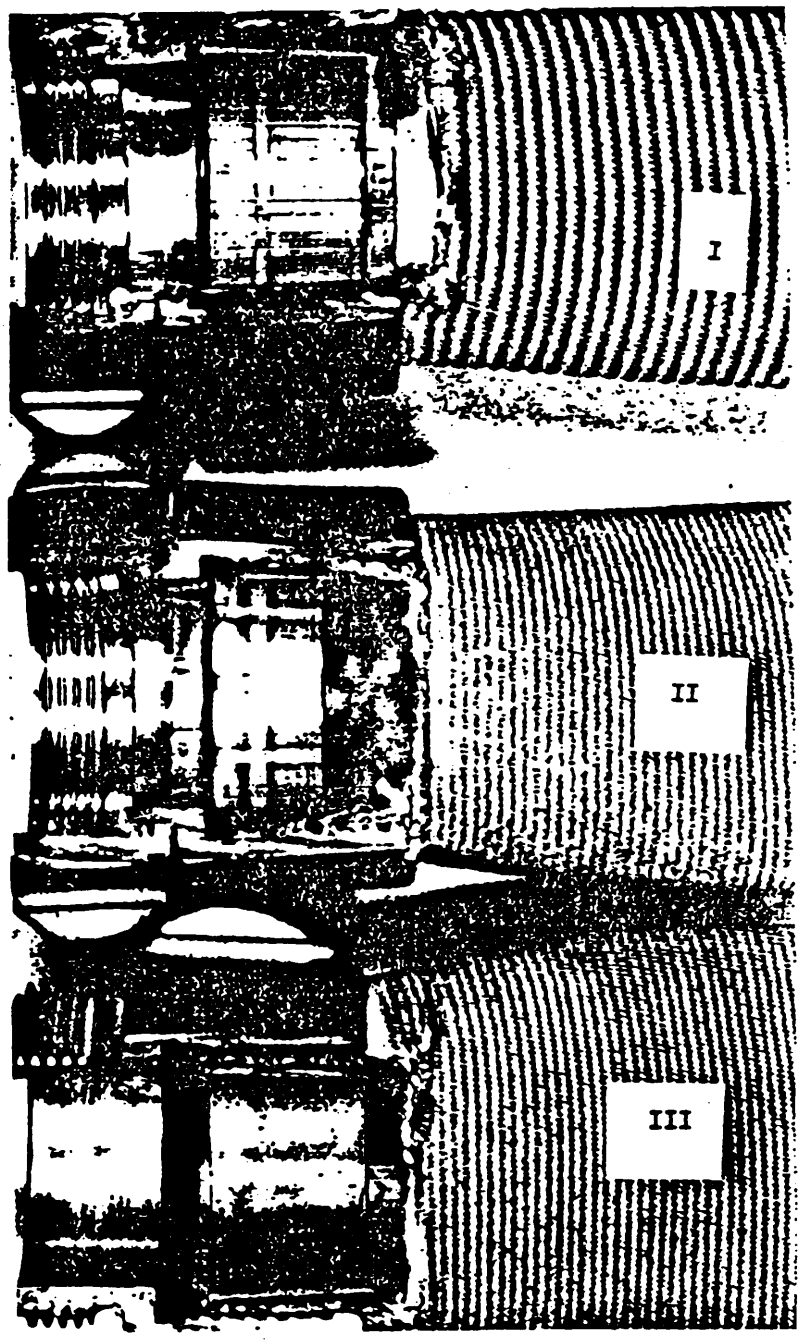


FIGURE 112. Ring, expansion, improperly placed:
I—Correct.
II—Ring in waterway—major.
III—Too much gap—major.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.



FIGURE 113. Cement in tube—minor.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

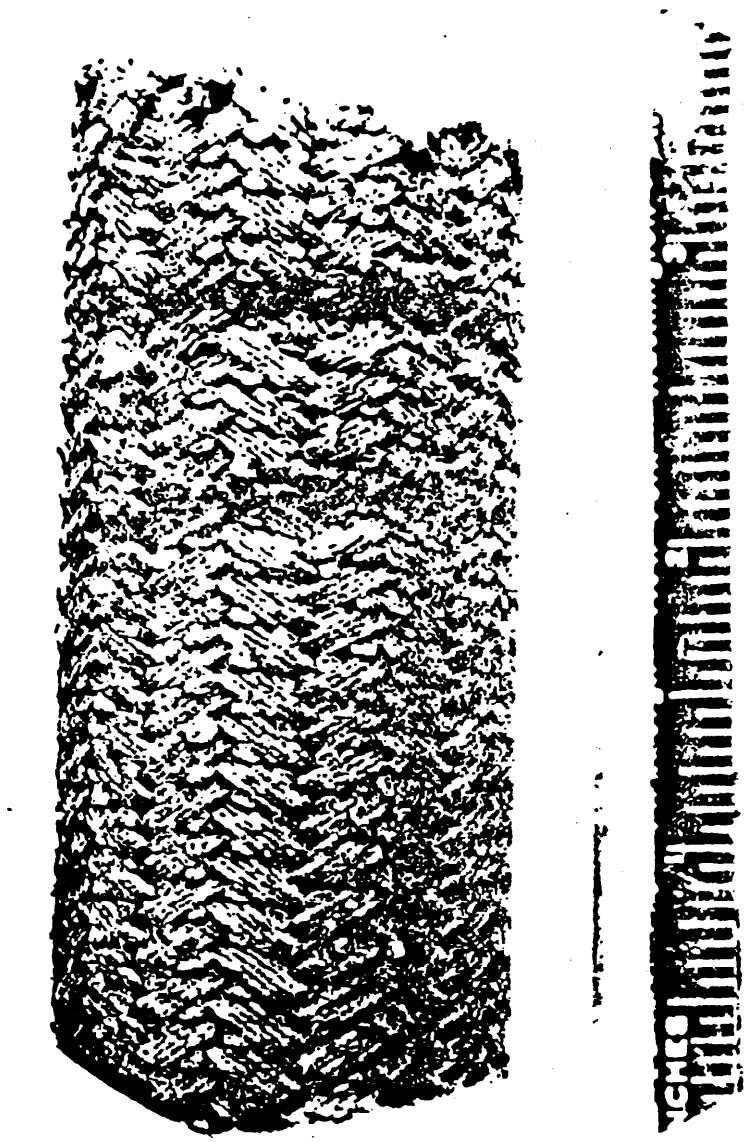


FIGURE 116. Clamp mark—major.
(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

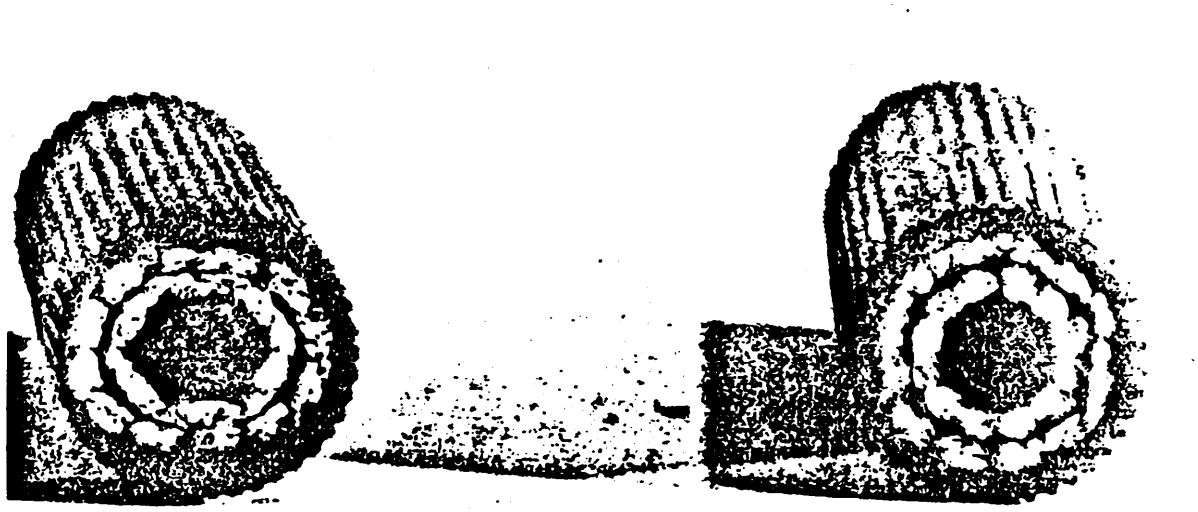
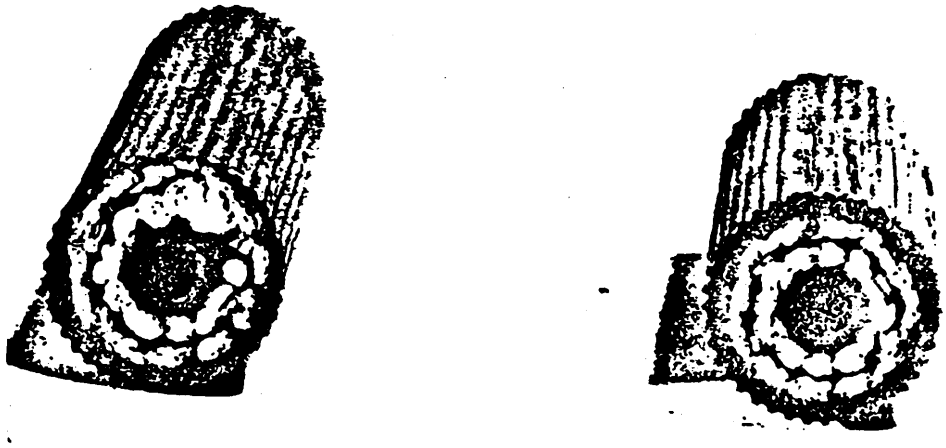


FIGURE 117. Clamp mark—major.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.



I II

FIGURE 130 Ball-up:
I—Yarn ba1l-up—major.
II—Normal hose.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026t0-26T12:54Z
Check the source to verify that this is the currentversion before use.

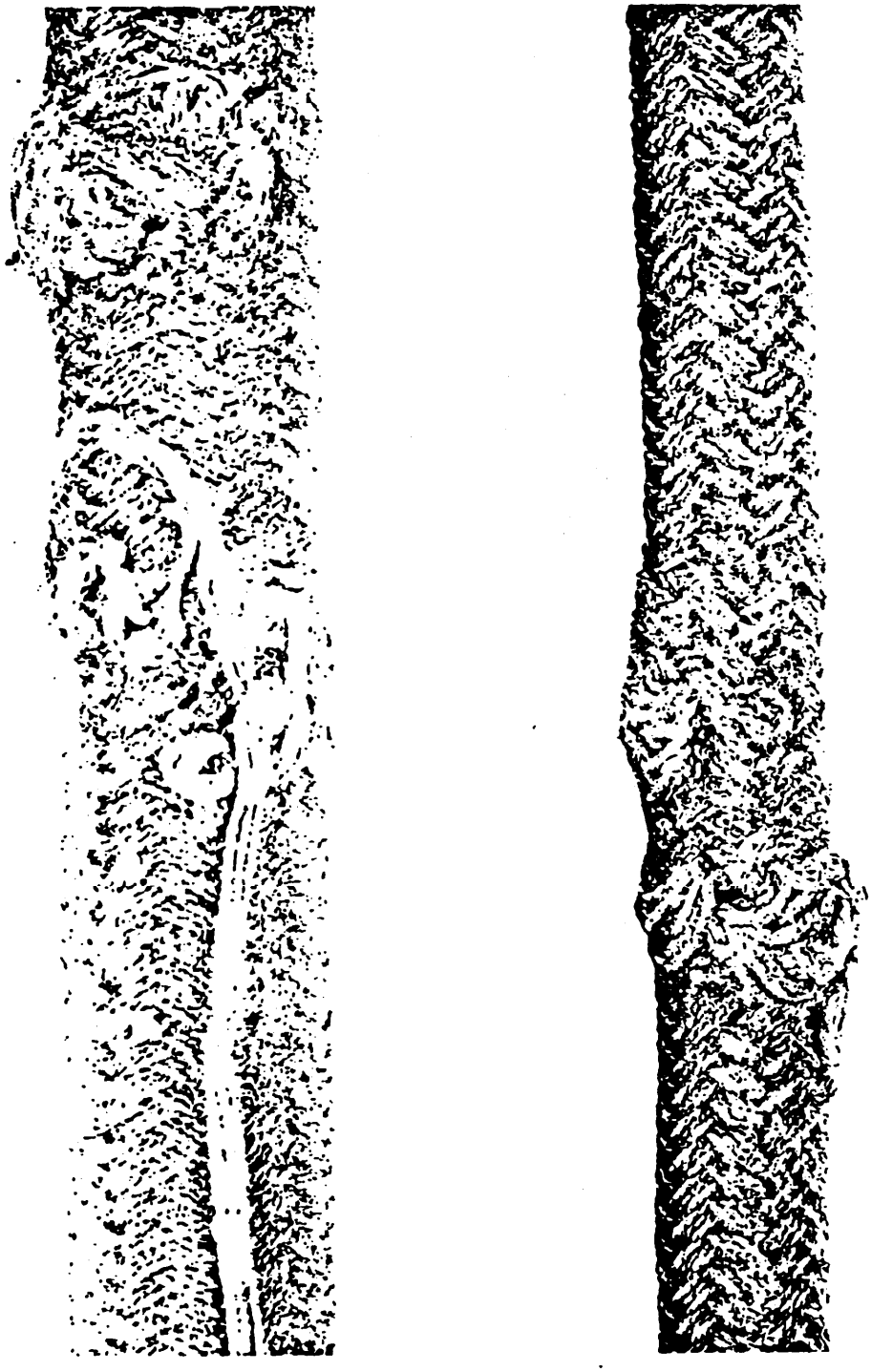


FIGURE 131. Ball-up—major.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

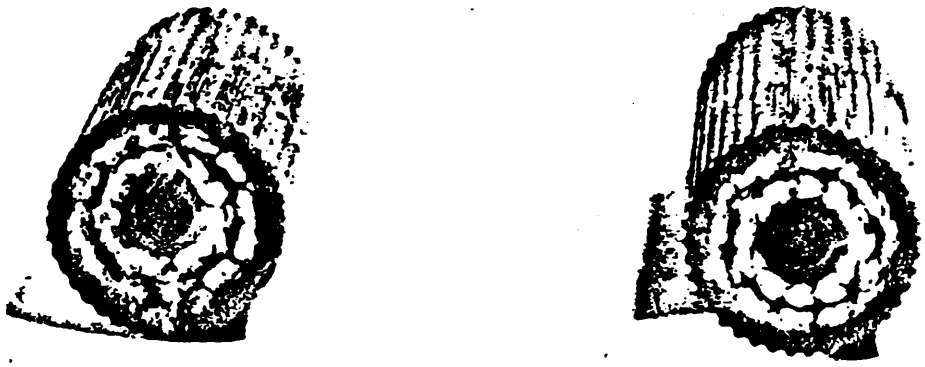


I II

FIGURE 138. Friction between braids, where specified, missing:
I—Friction missing—major.
II—Normal hose.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.



I II

FIGURE 139. Knot in yarn:
I—Knot in yarn—major.
II—Normal hose.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

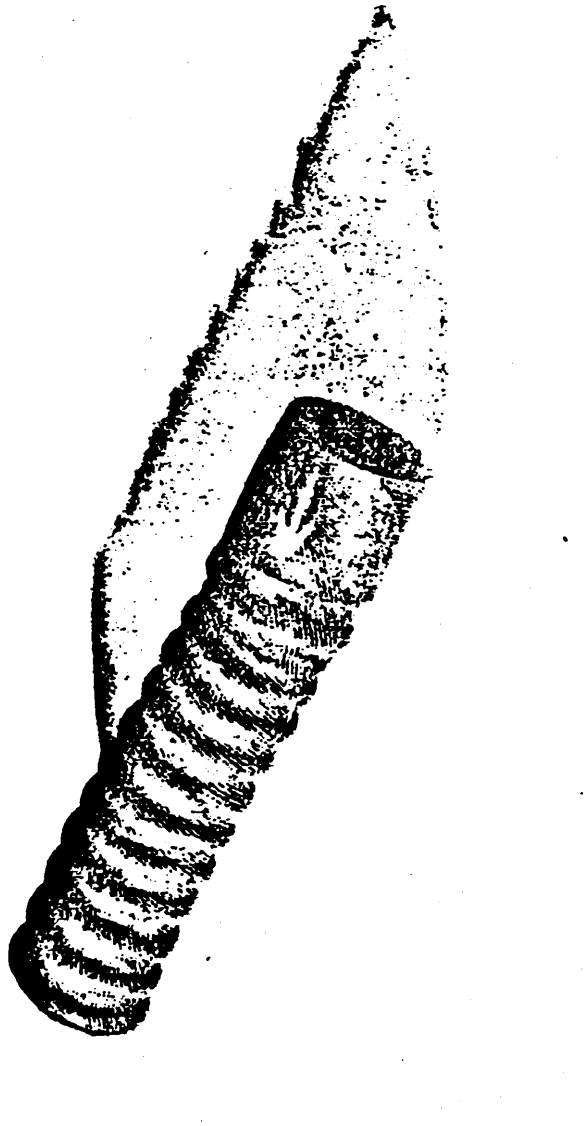


FIGURE 152. Wire through cover—major.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

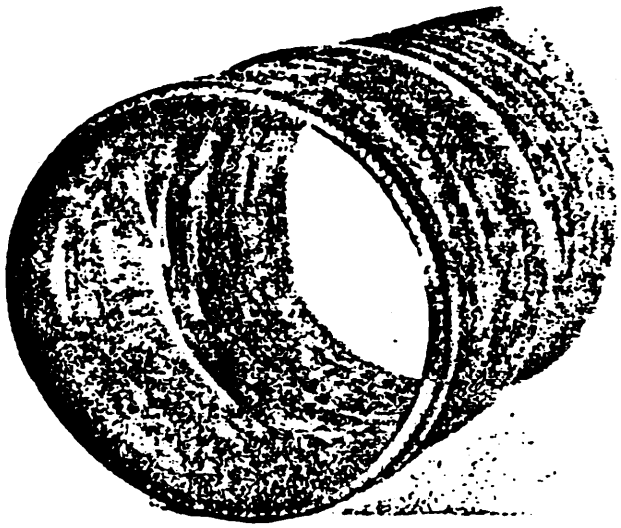


FIGURE 153. Wire through tube—major.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

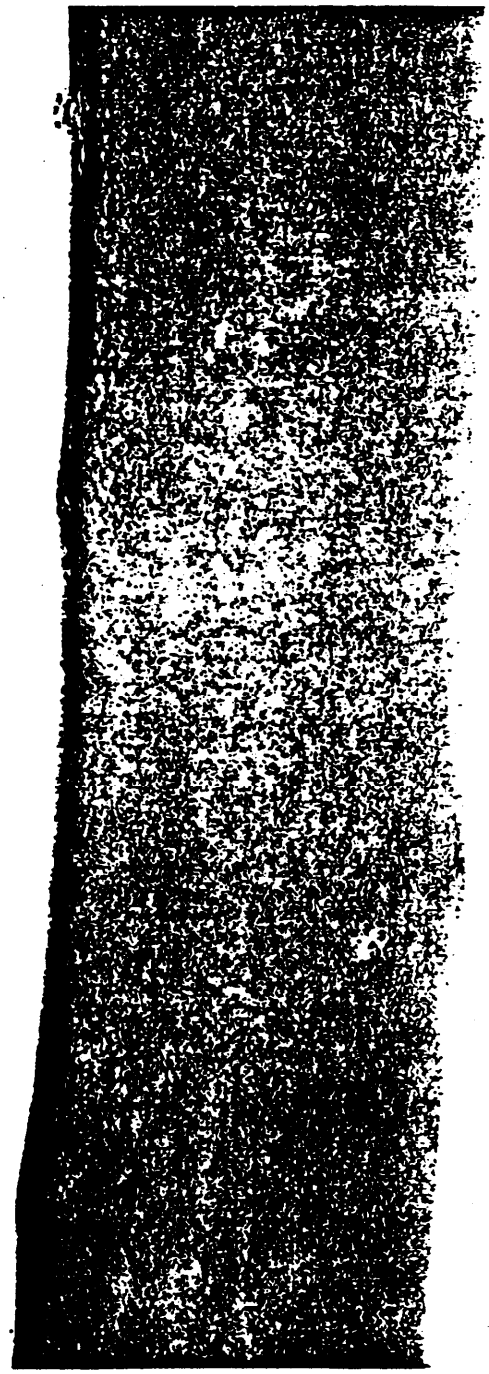


FIGURE 166. Wire, static, bulge: with loose cover—major; dynamic use (Air Force procurement only)—major; with adhered cover—not a defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

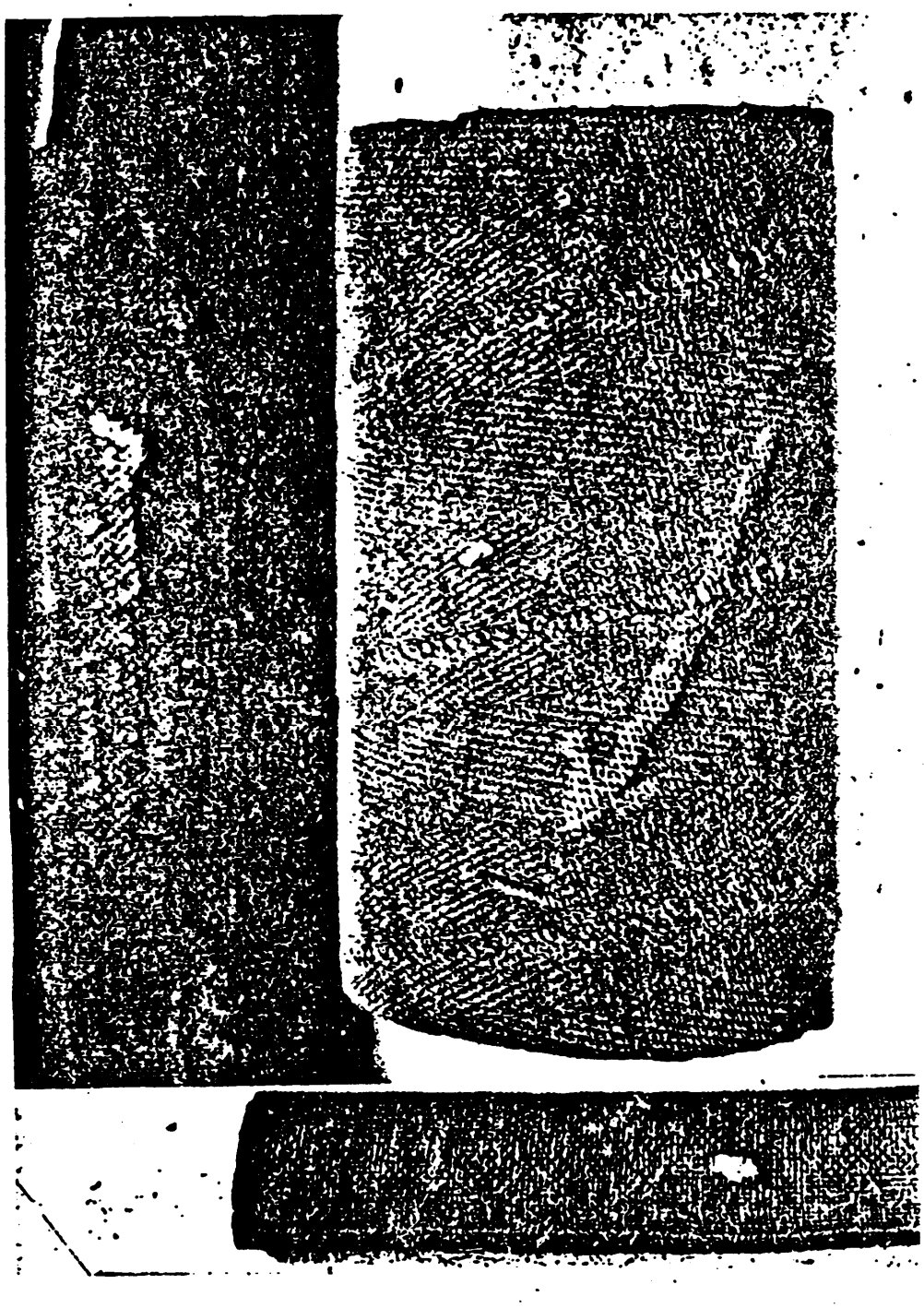


FIGURE 167. Yarn through cover—major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

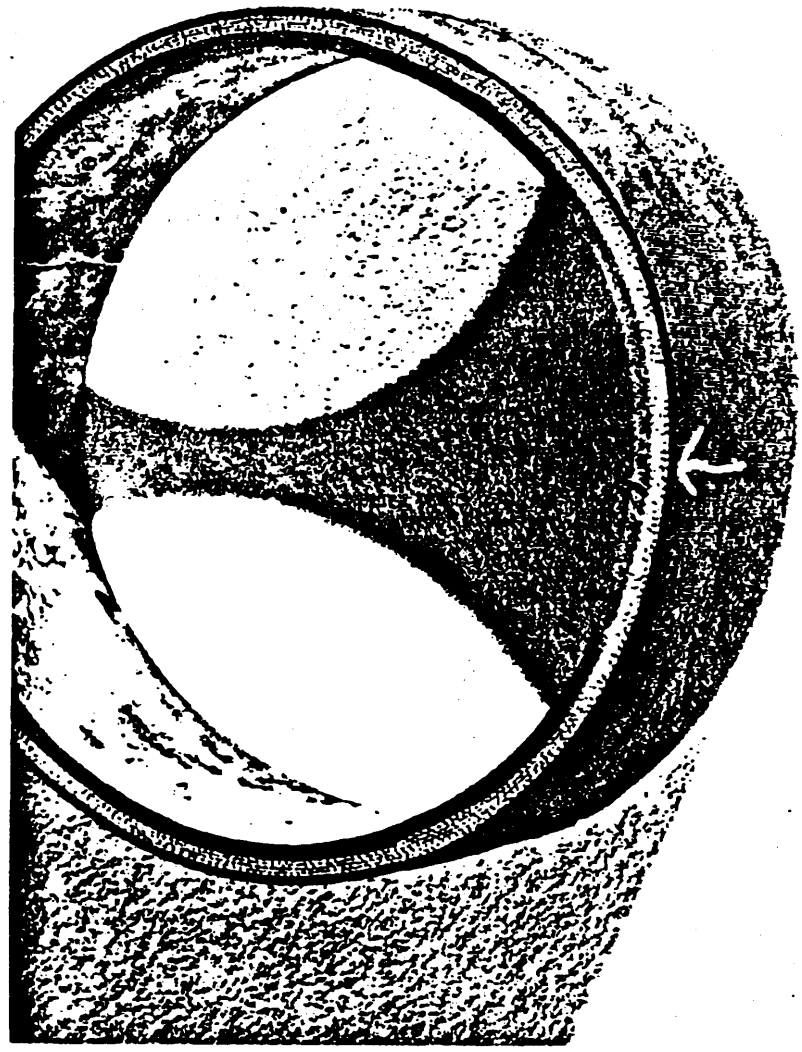


FIGURE 174. Hole—major.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

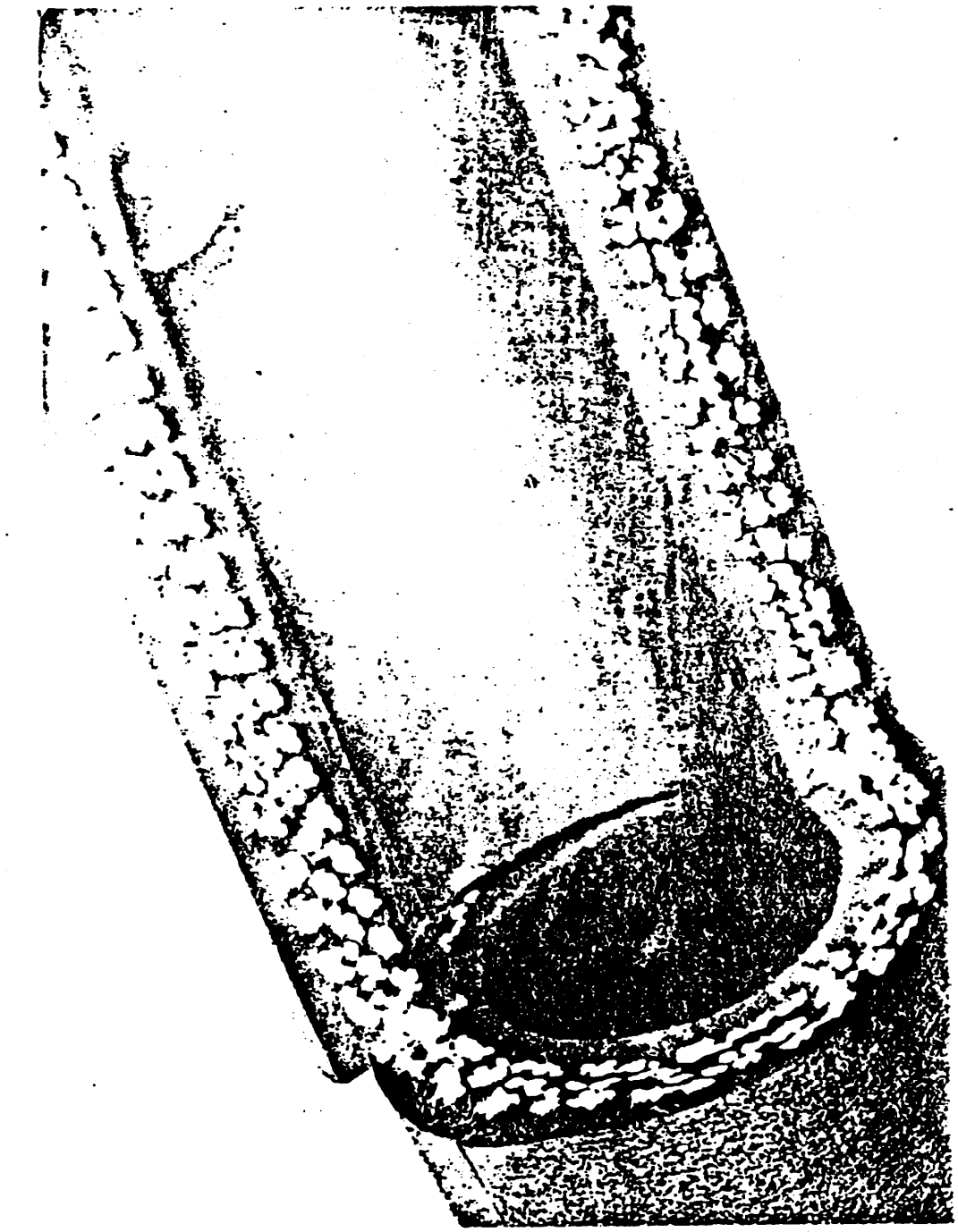


FIGURE 175. Hole—major.
(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

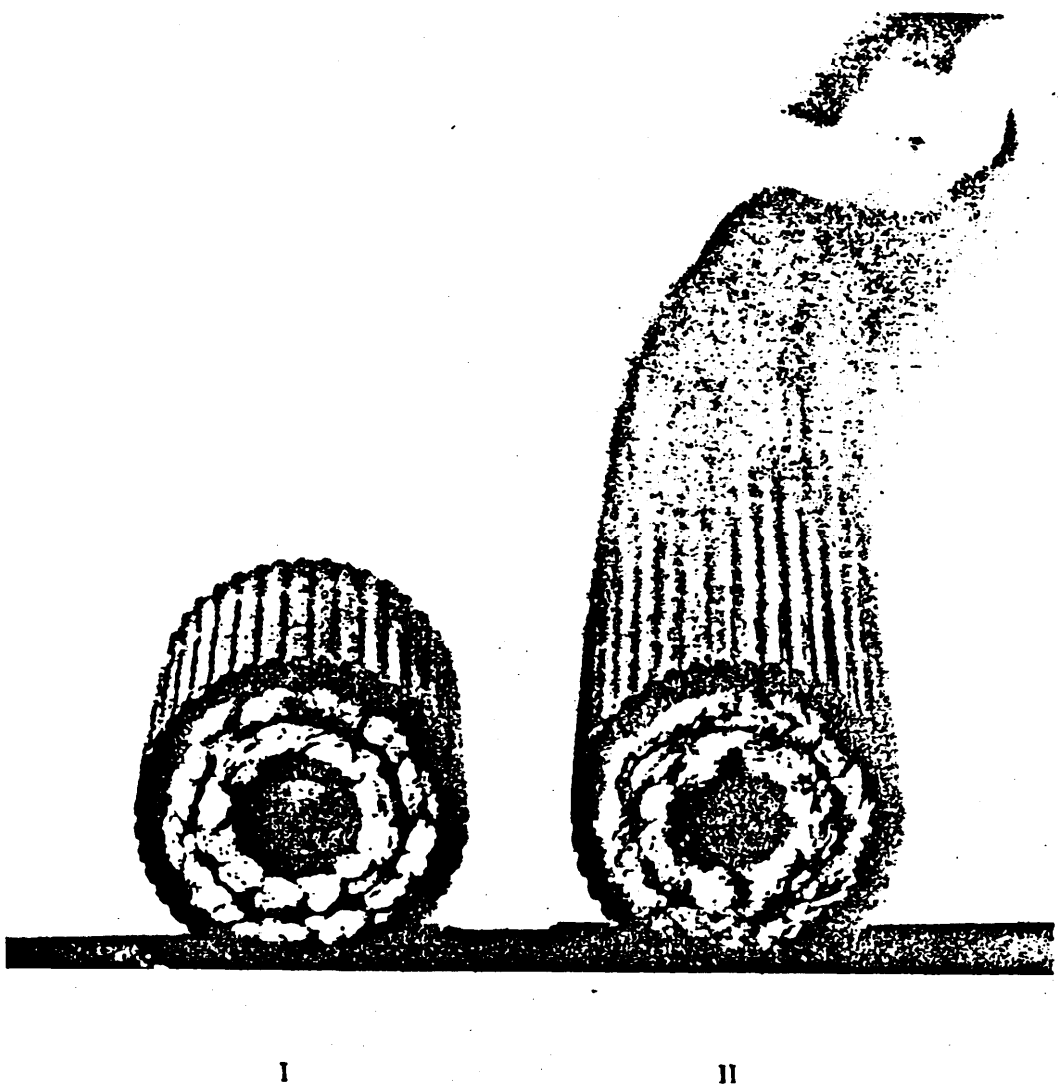


FIGURE 186. Tube missing:
I—Normal hose.
II—Major.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

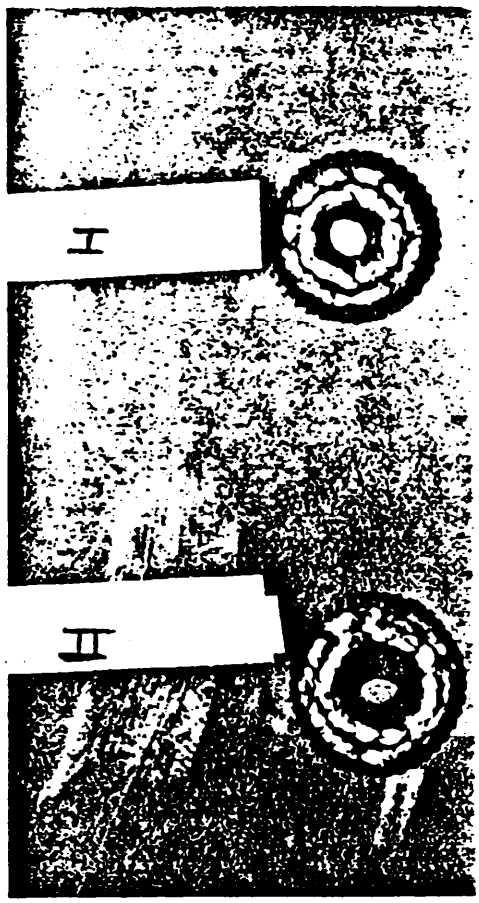


FIGURE 187. Off-center tube:
I—Thickness meets tolerance—not a defect.
II—Thickness below tolerance—major.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

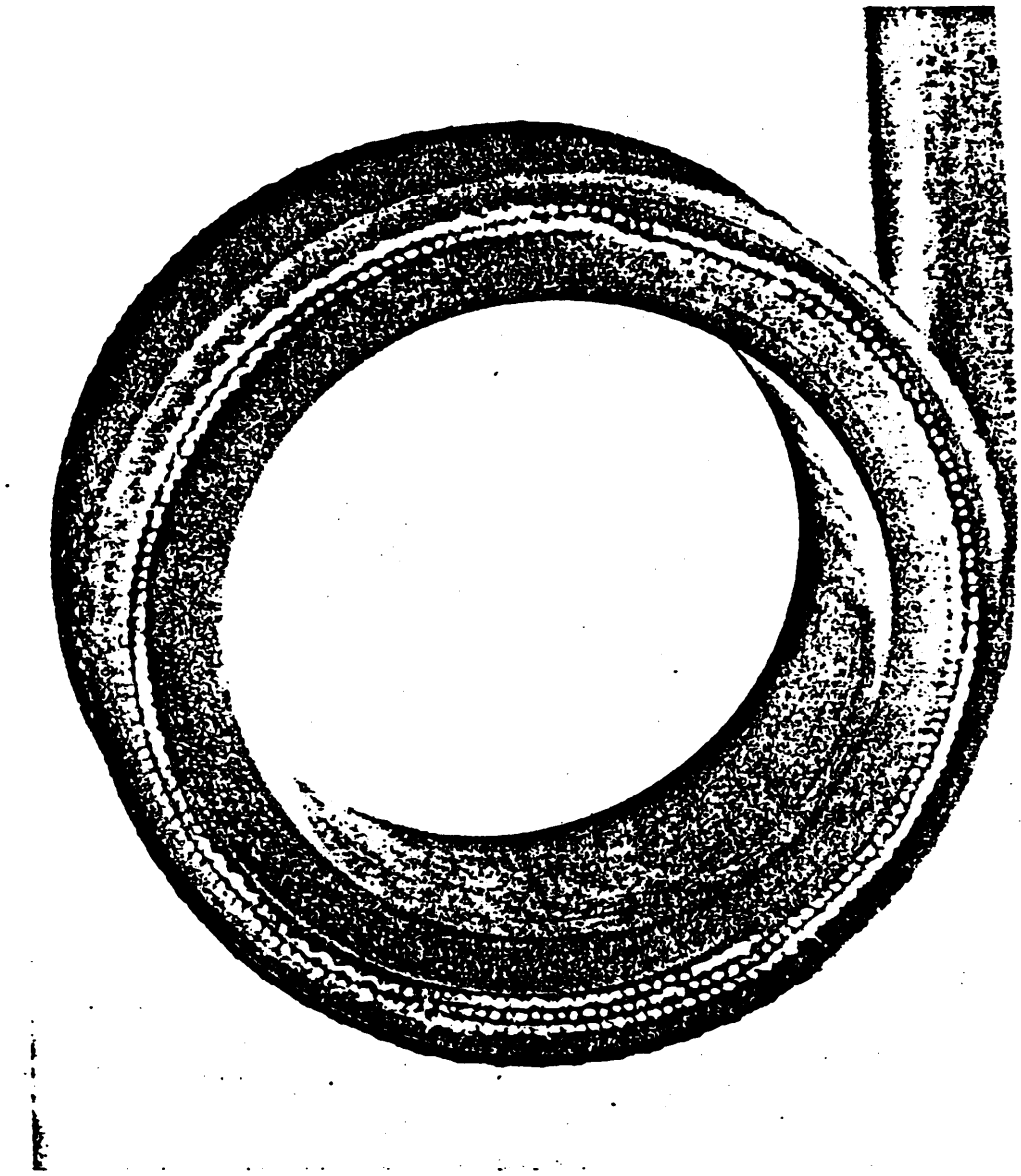


FIGURE 192. Porosity, more than 15 percent of tube thickness—major.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

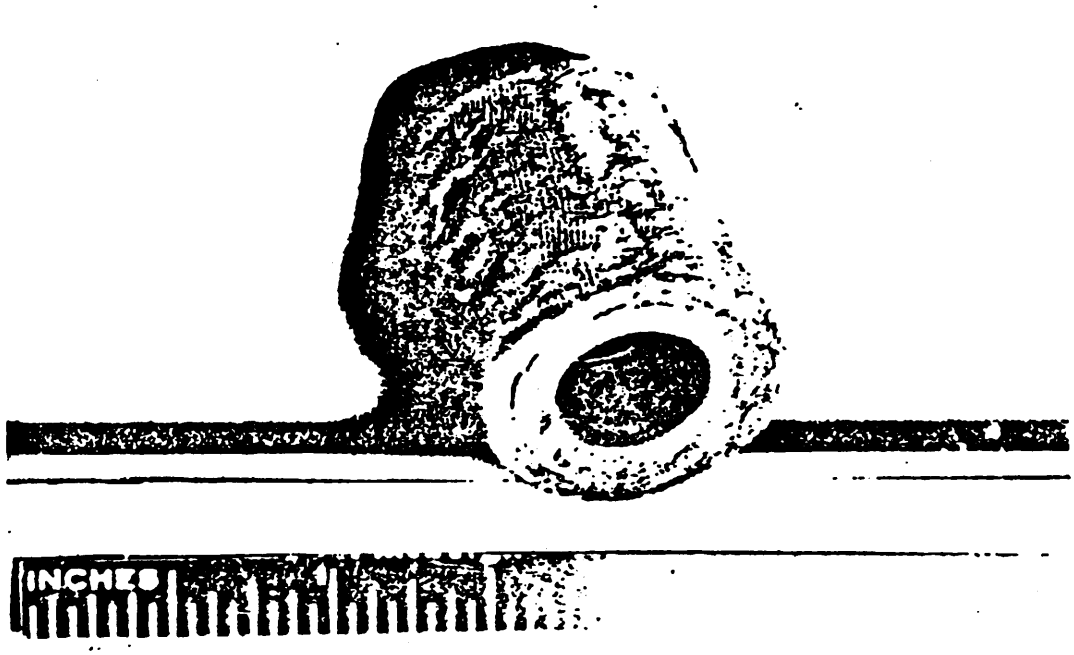


FIGURE 193. Ring: class A—major; classes B and C—not a defect.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

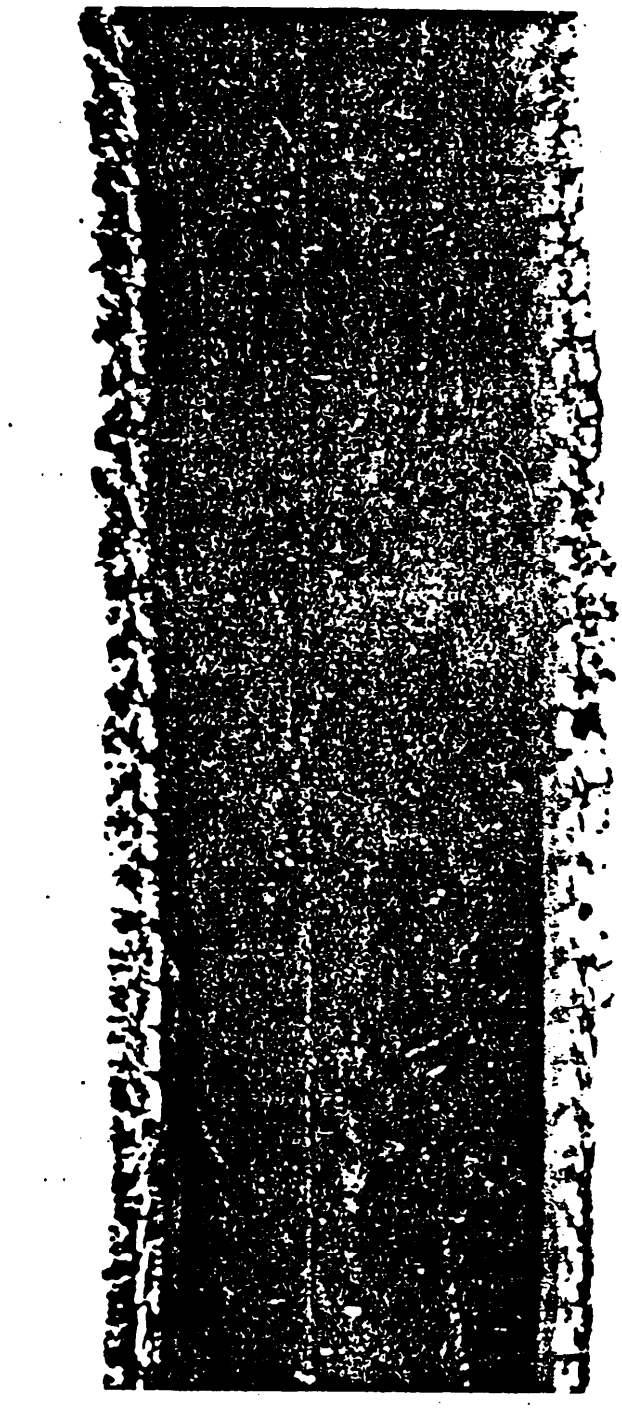


FIGURE 200. Surface irregularity: when assembled with swaged fitting—major; all others—minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

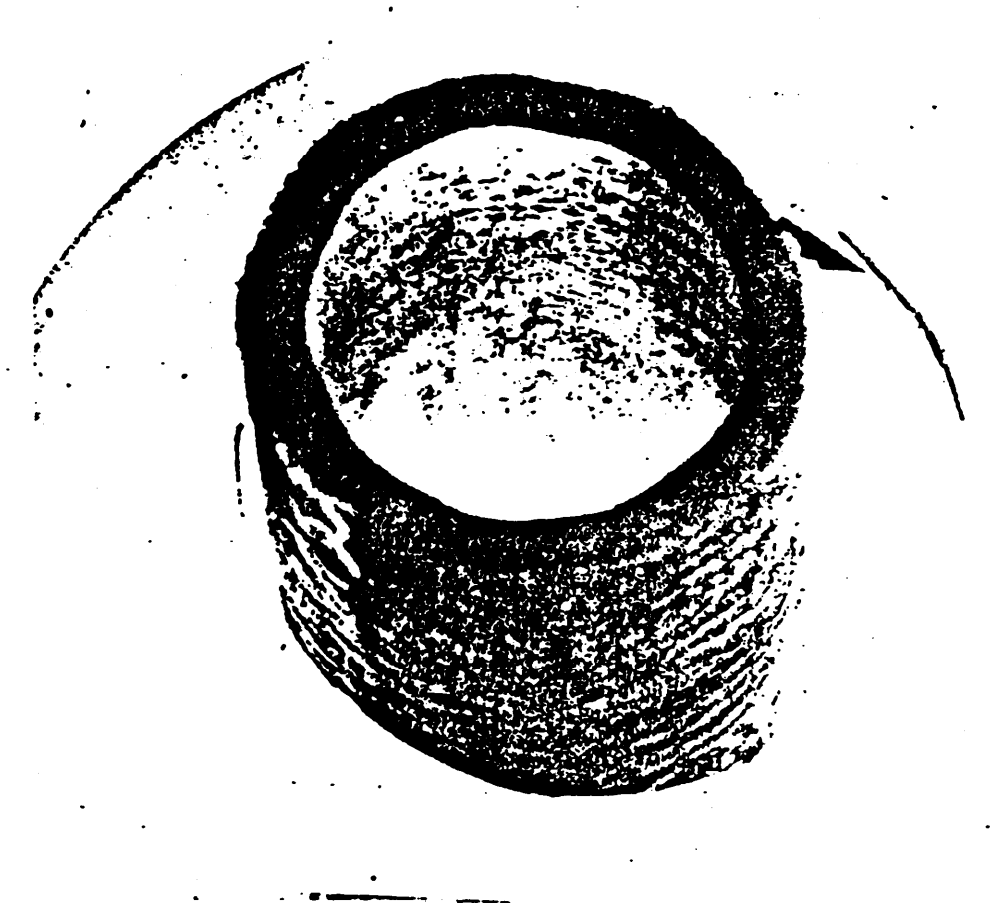


FIGURE 201, Surface irregularity: when assembled with swaged fittings—major; all others—minor.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

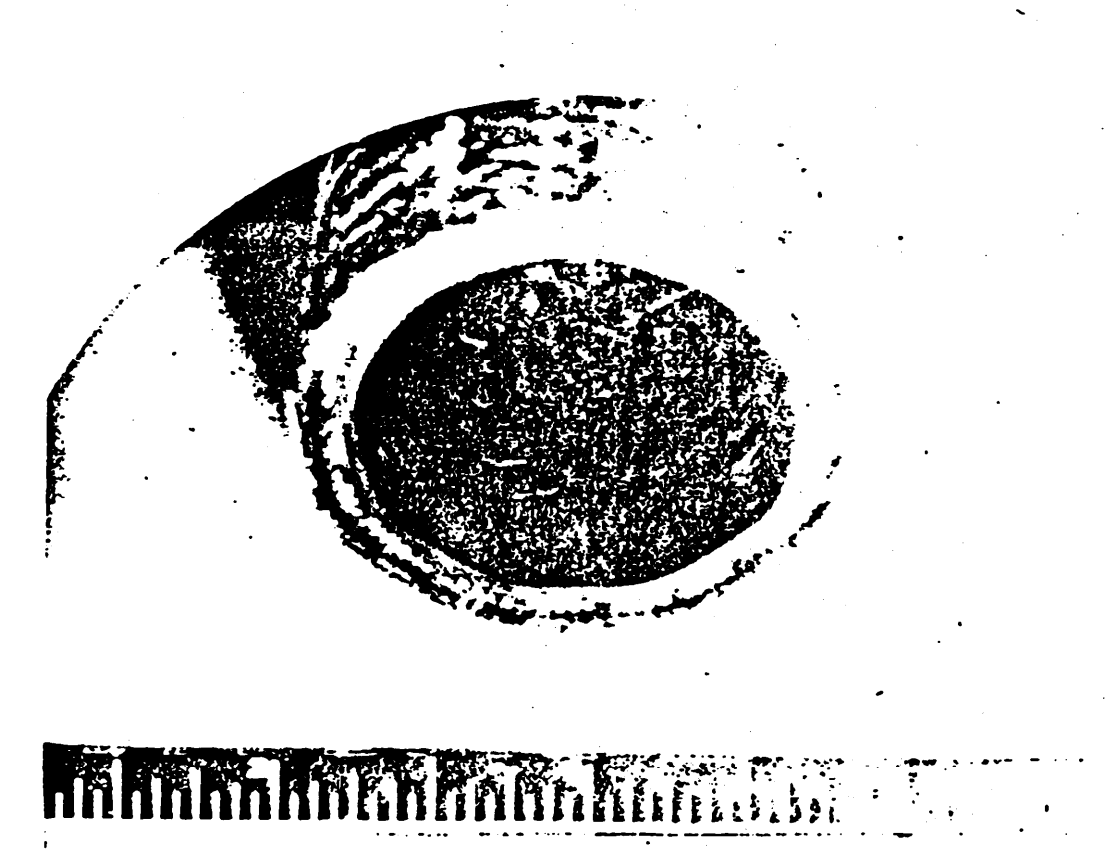


FIGURE 202. Surface irregularity: when assembled with swaged fittings—major; all others—minor.

(Corrected January 4, 1968)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.

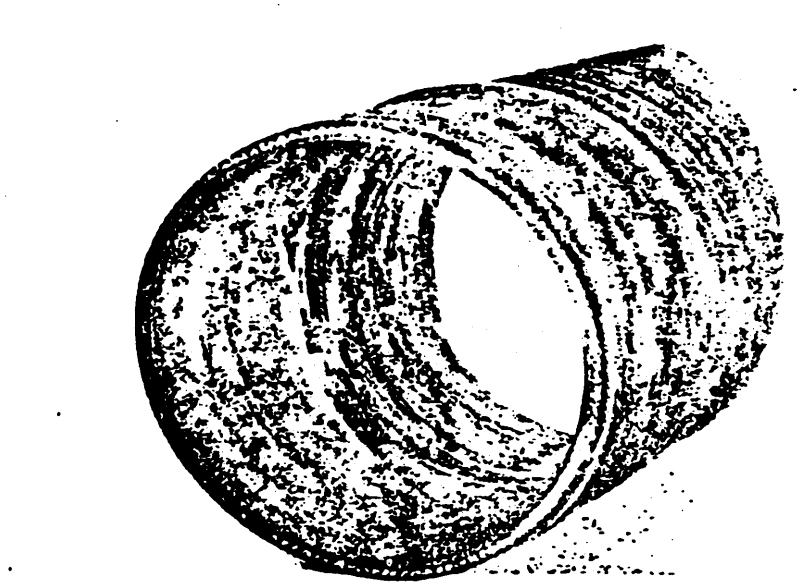


FIGURE 203. Wire through tube—major.

(Corrected January 4, 1968)

* U. S. GOVERNMENT PRINTING OFFICE: 1968 0 - 302-507 (514)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:54Z
Check the source to verify that this is the current version before use.